MURALI SAI BUDDAKKAGARI VENKATA

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EDUCATION

Stevens Institute of Technology **May 2026**
Master of Science, Applied Artificial Intelligence *NJ, US*

PES University **May 2024**
Bachelor of Technology, Computer Science (Specialization in ML) *KA, IND*

TECHNICAL SKILLS

Programming Languages: Python, **SQL**, **PySpark**

AI & Machine Learning: LLM Application Development, **Agentic Systems** & **Tool Orchestration** (**MCP**), **Retrieval-Augmented Generation** (Chunking, Reranking, Grounding), **Prompt Engineering** (Chain-of-Thought, Few-Shot), **Function Calling**, **Multi-Agent Systems**, **Agent Evaluation** (RAGAS), Guardrails & Reliability, **Multimodal AI**, **NLP**, **Embeddings**, **Vector Databases**, **LangChain**, **LangGraph**, **PyTorch**, **TensorFlow**, **HuggingFace**

Infrastructure & DevOps: FastAPI, **ChromaDB**, **Docker**, OpenTelemetry (Observability), **Google Cloud Platform**, **Azure**, **AWS** (Glue, S3, Redshift), PostgreSQL, **GitHub Actions**, **CI/CD**, **Git**, **REST APIs**, ReactJS, **OpenAI API**, **Claude API**, **Gemini API**

EXPERIENCE

Querkey Inc **Jun 2025 - Aug 2025**
Data Science Intern *CA, US*

- Developed and deployed a proof-of-concept real-time Patient Flow Tracking System using **multimodal AI** (Gemini 2.5 Flash) with **chain-of-thought** and **few-shot prompting** to improve routing efficiency, integrating a **RAG** pipeline that structured unstructured patient notes for real-time clinical insight retrieval.
- Cut production **ML** workflow failures 60% across 100K+ records by deploying a **Dockerized** observability stack (**MLflow**, **FastAPI**, **ChromaDB** vector store, **HuggingFace** monitoring), enabling real-time failure detection.
- Built OLTP-to-OLAP data pipelines using **PySpark** and **AWS Glue**, transforming and loading logistics operations data into Amazon S3 and Redshift for downstream analytics.

VIEW GROUP INDIA (FORMERLY VIEW SYNERGY) **Jan 2024 - Apr 2024**
AI and Automation Intern *KA, IND*

- As part of a **cross-functional** project team modernizing a live production platform, reviewed **React** frontend, .NET **Azure** Functions **API**, and **MySQL** stored-procedure code, tracing 2 recurring exceptions to root cause and surfacing **SQL**-injection vulnerabilities that fed into a 94-issue security assessment.
- Built and tested a new Roles/Designations **CRUD** feature end to end, including new **MySQL** stored procedures, a .NET **API** layer, and a **React** frontend, while documenting **Microsoft Entra ID** role setup for a proposed identity migration.
- Audited **Redis** usage across all 6 backend microservices and documented **Microsoft Entra ID** app registration and role setup, supporting a proposal to replace a fully custom **JWT**/**BCrypt** authentication system.

PROJECTS

SEC EDGAR RAG Filing Analyzer

- Deployed a production **RAG** system chunking and indexing 8,232 **SEC 10-K** filing chunks across 5 companies, verified by a 504-test regression suite run in **CI** and a 54-question **eval** suite (0.853 faithfulness, 0.936 citation accuracy, 0.981 refusal correctness), by building a **FastAPI** + **ChromaDB embeddings** pipeline with **cross-encoder reranking**, **RBAC**-enforced information barriers (FINRA Rule 2241 / SEC Rule 15g-1), and a serverless **Google Cloud Run** deployment (scale-to-zero).
- Built literal **entity detection** pinning retrieval to the named company's filing at the database level, per-entity retrieval budgets for comparative questions, and a bounded **agentic** retry with **prompt-injection** and MNPI/ investment-advice output **guardrails**.

Agent Orchestration System with Tool Use, Memory, and Human in the Loop

- Shipped a production **multi-agent** system (19 **REST** endpoints) that reasons, plans, and executes multi-step tasks end-to-end, validated by 106/113 passing tests against a live deployment, by architecting a 3-layer **LangGraph** hierarchy (planning supervisor, parallel specialists, reflective reviewer) with multi-model routing (GPT-4o, Claude Haiku) and two-tier **Redis** + **Supabase pgvector** memory.
- Built a custom **MCP** (Model Context Protocol) server integrating internal and third-party tools via **APIs** across 6 tools, instrumented end to end with **OpenTelemetry** distributed tracing, and a **Human-in-the-Loop** system spanning 4 escalation severity levels with full audit-trail logging.

PUBLICATIONS AND CERTIFICATIONS

- Adaptive Ensemble Detection for Medical Imaging, M.S. Capstone, Stevens Institute of Technology (Advised by Prof. Kevin Lu). A size-conditioned Weighted Box Fusion ensemble of Faster R-CNN, Mask R-CNN, and YOLO26 for cell-nucleus detection, reaching 0.71 mAP@0.5 on the 2018 Data Science Bowl benchmark.
- Best Research Paper Award, ICICC 2024 - WorkFit: Real-time Multimodal Posture Monitoring System using CNN-based Computer Vision and Flex Sensor Fusion.